



Midgard Project Close-Out Report

Project Catalyst Final Milestone Submission

Project: Midgard: Cardano Layer 2

Project ID: 1200027

Fund: Fund 12

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Executive Summary

Midgard is a Cardano Layer 2 protocol designed to improve throughput, reduce congestion pressure on Cardano Layer 1, and support faster execution for applications that need higher-frequency transaction handling. The project focused on delivering an optimistic rollup-style protocol with on-chain Cardano contracts, an off-chain SDK, a demo node runtime, Docker deployment infrastructure, and public technical documentation.

The funded work produced the core protocol architecture, smart contract suite, state management mechanisms, inbox and outbox user-event contracts, and a node MVP capable of interacting with Cardano preprod. The final MVP evidence shows protocol initialization, operator registration and activation, state commitments, state merge, deposit handling, withdrawal ordering, and payout settlement on preprod.

Completed Project Outputs

- Technical architecture and protocol specification for the Midgard Layer 2 design.
- On-chain Aiken validators and supporting protocol libraries.
- State management contracts for state commitments and settlement.
- Inbox and outbox contracts for deposits, transaction orders, withdrawals, and settlement.
- Off-chain SDK and transaction-building support.
- Demo node/runtime implementation with API routes, workers, services, and command endpoints.
- Docker infrastructure for local deployment and node management.
- Public preprod transaction evidence demonstrating the MVP protocol lifecycle.
- Final close-out video demonstrating the Midgard MVP and reflecting on the work completed.

Public Evidence

Primary repository:

<https://github.com/Anastasia-Labs/midgard/tree/tx-validation>

Pinned implementation evidence used for final milestone review:

<https://github.com/Anastasia-Labs/midgard/tree/367ede9ae356ef4da0f0fb45b7eedc59c64585d6>

Technical specification:

<https://anastasia-labs.github.io/midgard/midgard.pdf>

Final close-out video:

<https://www.youtube.com/watch?v=I8G4bFIBN5Q>

MVP Demonstration Evidence

The MVP was exercised on Cardano preprod through the published Midgard contracts and node/runtime flow. Key evidence includes:

- Protocol initialization:
<https://preprod.cardanoscan.io/transaction?id=47d17f1b0f93b61bdfccbe6623166ed6132077d7222f7afcb9144c70048be41a>
- Operator registration:
<https://preprod.cardanoscan.io/transaction?id=288e1a15e1145a91d35ba3729c0f5c7c73cd5535061830632ef2c30c2b5769ea>
- Operator activation:
<https://preprod.cardanoscan.io/transaction?id=a83e282e09bf61019baa330ed7a63d37d4272281ccfec864111f9879022197a>
- Deposit transaction:
<https://preprod.cardanoscan.io/transaction?id=637291210692b6b00edf9f09acf11a8fdd25c75eca4115c0f0ec718676c5528a>
- State commitment 1:
<https://preprod.cardanoscan.io/transaction?id=fa51a5ac0e28776d0cde3aa9c57f345346e385531bc7fd0fff991e799032e845>
- State commitment 2:
<https://preprod.cardanoscan.io/transaction?id=0c5e43e57b472493b54bfe2dfcf4a424ceeb0ca38a82342ef8f83504b54c733b>
- State merge:
<https://preprod.cardanoscan.io/transaction?id=22e6b1e0a1f19c064f67b0b6ce1af03293bc488bf653b7f255feaf38752df810>
- Withdrawal order:
<https://preprod.cardanoscan.io/transaction?id=6c34f10740f728cb8618639480c3dd1cfc55a88bd4069395548741121b1c7534>
- Payout initialization:
<https://preprod.cardanoscan.io/transaction?id=8c15f705e2ce50147eb42c53a982c9556f44118c80aad3da28f8aaa6e0c524f9>

- Reserve funding for payout:
<https://preprod.cardanoscan.io/transaction?id=db99e44efb291abc3ca9c376ef7f9ce91d73ef22380d236ae77571eb856fea93>
- Final payout settlement:
<https://preprod.cardanoscan.io/transaction?id=ca8dd57556bfb7a0fe60d3f71ace9bffe6e176d9e8805f272127e13dceba85a>

Retrospective

The project progressed from a technical architecture and protocol design effort into a working public MVP. The most important achievement was tying together the on-chain contract suite and off-chain runtime into an end-to-end preprod flow. That flow demonstrates the core Midgard concept: users interact with Layer 1 contracts, the node observes and processes events, state commitments are published, and settlement or withdrawal actions can be verified against the published protocol state.

The work also clarified which parts of the system are stable enough for public review and which parts should continue as follow-on engineering work. The MVP demonstrates the essential protocol lifecycle, while future production hardening should focus on broader operator participation, additional testing, performance measurement, monitoring, and project onboarding.

Challenges Encountered

The main challenge was coordinating a protocol that spans multiple layers at once: Cardano Layer 1 validators, off-chain transaction construction, protocol state management, node workers, database persistence, API services, and deployment infrastructure. Progress required keeping the technical specification, implementation, and preprod evidence aligned as the design matured.

Another challenge was producing reviewer-friendly public evidence for a complex protocol. The project contains many separate components, so the final milestone evidence needed to connect source code, documentation, Docker/runtime instructions, and on-chain transaction evidence into a coherent review path.

The project also required careful iteration around the MVP scope. Some longer-term goals, such as production-scale multi-operator deployment and onboarding additional projects, remain natural next steps beyond the MVP evidence submitted here.

Key Lessons Learned

The first key lesson is that an L2 protocol needs evidence at multiple levels. Source code is necessary, but not sufficient by itself. Reviewers and users also need build instructions, runtime documentation, and on-chain execution evidence that shows the components working together.

The second lesson is that protocol documentation must stay close to implementation. Midgard's technical specification, Aiken validators, SDK, and node runtime are most useful when they can be reviewed together rather than as disconnected artifacts.

The third lesson is that MVP delivery benefits from a focused demonstration path. A smaller, well-evidenced preprod lifecycle is more useful for milestone review than a broad set of claims without transaction-level support.

Next Steps

- Continue hardening the node runtime and API surface.
- Expand automated testing and integration coverage.
- Extend the system toward broader multi-operator participation.
- Improve deployment and observability tooling.
- Prepare onboarding documentation for early projects that want to test Midgard.
- Continue publishing technical updates and implementation evidence in the public repository.

Close-Out Statement

The Midgard Catalyst project has delivered the core public artifacts required for the MVP: protocol documentation, open-source implementation, node/runtime tooling, deployment infrastructure, preprod transaction evidence, and a public close-out video. Together, these outputs show meaningful progress toward a Cardano Layer 2 protocol and provide a clear foundation for continued development beyond the funded milestone scope.